SURVEY

Lightweight Consensus in Blockchain: A Systematic Literature Review

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Open Access Support provided by:

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Published: 07 October 2025
Online AM: 15 September 2025
Accepted: 05 September 2025
Revised: 22 February 2025
Received: 30 May 2023

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Blockchain technology has seen a rapid pace of development and expanded application domains swiftly due to the rising demand for decentralized trust, transparency, and integrity. The consensus algorithm plays a critical role in ensuring trust, immutability and governance of the decentralized network. However, traditional consensus face challenges such as high energy consumption, low scalability, security, and fault tolerance. Researchers have been investigating Lightweight Consensus to overcome these challenges. Lightweight Consensus is a mechanism that enables a more efficient and scalable blockchain system while ensuring security and immutability. This work uses the Systematic Literature Review method to comprehend Lightweight Consensus. 127 studies were grouped based on application specific network, and an in-depth analysis was done on the characteristics of the consensus. A novel taxonomy of Lightweight Consensus based on the agreement method and round propagation is proposed. Various parameters that needed consideration for a Lightweight Consensus are also analyzed. Finally, the study makes recommendations for future research on Lightweight Consensus in blockchain, emphasizing the importance of more empirical investigations and real-world implementations. This study offers a comprehensive overview of the current research landscape on lightweight consensus in blockchain, shedding light on its potential impact on the evolution of blockchain technology. It also serves as a valuable guide for researchers, helping them identify the most suitable consensus features for specific application domains with unique requirements.

CCS Concepts: • **Networks** → *Network types*; Peer-to-Peer protocols; • **Computer systems organization** → *Peer-to-Peer architectures*; *Reliability*; • **Computing methodologies** → Distributed algorithms; • **Security and privacy** → *Cryptography*; • **General and reference** → **Surveys and overviews;**

Additional Key Words and Phrases: Blockchain, lightweight consensus, networks, performance

ACM Reference Format:
Namrata Marium Chacko, Narendra V G, Mamatha Balachandra, and Manoj T. 2025. Lightweight Consensus in Blockchain: A Systematic Literature Review. *ACM Comput. Surv.* 58, 3, Article 78 (October 2025), 37 pages. https://doi.org/10.1145/3768149

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ACM 0360-0300/2025/10-ART78
https://doi.org/10.1145/3768149

1 Introduction

In the past decade, there has been prolific advancement in blockchain technology [8]. Blockchain offers a solution to various problems faced by centralized systems. Blockchain applications include decentralized finance, assets and non-fungible tokens, energy markets, health care, intelligent vehicular networks, supply chains, and precision agriculture. The popularity of blockchain is due to its ability to maintain the reliability and immutability of transactions via consensus algorithms in a trustless, decentralized environment.

The driving force behind blockchain is the consensus method. The consensus determines the reliability of the blockchain network [36, 80]. This, in turn, also reflects on the quality of service parameters of the network, like scalability, latency, security, and efficiency [127]. There are various consensus schemes; not all can be applied to all areas. Researchers have been working on making resource-efficient consensus protocols and termed them lightweight or ultra-lightweight.

1.1 Challenges of Traditional Consensus

Traditional consensus methods such as **Proof of Work (PoW)** and **Proof of Stake** (**PoS**) have been widely embraced in blockchain technology for confirming transactions and maintaining the blockchain's integrity. However, these algorithms have encountered several difficulties. The major challenges faced are outlined below [7]:

(1) **Scalability:** The number of nodes in a blockchain network can affect the performance of the consensus mechanism, leading to slower transaction processing times.
(2) **Security:** Consensus methods must ensure that transactions are accurately authenticated and that hostile actors cannot influence the network's state.
(3) **Energy consumption:** Some consensus methods need a large amount of processing power, which might result in excessive energy consumption and environmental problems.
(4) **Fault tolerance:** Consensus techniques must be capable of dealing with node failures or malicious behavior without jeopardizing network security or performance.
(5) **Privacy:** Some consensus methods may expose sensitive information about network transactions or participants, which may worry privacy-conscious users.
(6) **Interoperability:** Different blockchain networks may utilize different consensus techniques, making it difficult to interoperate.

These issues underline the importance of additional research and the development of consensus algorithms capable of resolving these concerns while maintaining high performance and security requirements.

1.2 Organization of the Paper

The article is organized into the following sections. The main contributions are summarized in Section 2. Section 3 outlines the research questions and the detailed methodology used for data collection and the systematic literature process. Also, the bibliometric analysis is presented in this section. The scope of this article and the evolution of Lightweight Consensus algorithms through a timeline diagram is depicted in Section 4. The various domains authors consider to achieve Lightweight Consensus are outlined in Section 5. The performance of the lightweight is discussed in Section 7. The next Section 9, conjecture to research question, attempts to answer the research questions. Finally, Section 11 discusses the findings, conclusion, and limitations of this study.

2 Summary of Main Contributions

Despite various surveys and reviews of articles on blockchain, there has never been a systematic review of Lightweight Consensus and its impact on various applications regarding resource efficiency and performance capability. The significant contributions of this article are highlighted below.

Table 1. Summary of Prior Surveys

Reference	Article's Contribution	Our Contribution
Bahareh Lashkari & Petr Musilek [61]	Survey of 130 existing consensus	+Specific Lightweight Consensus.
Arshdeep Singh et al. [93]	Analysis of 50 consensus protocols based on parameters.	+Parameters essential for Lightweight Consensus.
Jeff Nijsse & Alan Litchfield [82]	A taxonomy and classification of blockchain consensus in existing blockchain.	+ A novel taxonomy specific to Lightweight Consensus
Sarah Bouraga [12]	Categorization methodology for various consensus mechanisms used in blockchain.	+ Our categorization is based on the application specific network.
Wei Yao et al. [122]	The research thoroughly examines 8 consortium blockchain consensus algorithms.	+ We explore all types of blockchain algorithms.
Umesh Bodkhe et al. [11]	This reviews consensus employed in cyber-physical systems applications.	+ We explore all applications.

(1) Taxonomy and categorization of Lightweight Consensus protocols.
(2) Correlate the evolution of consensus concerning issues faced by evolving blockchain applications.
(3) A parametric analysis of the impact of the consensus protocol on the application specific network,[1] the challenges, and possible solutions.
(4) Discussion of the basic characteristics of Lightweight Consensus.
(5) List the properties of Lightweight Consensus.
(6) Discussions on future directions of Lightweight Consensus protocols in distributed systems.

Table 1 summarises survey/review publications on blockchain consensus. It also highlights the differences between prior reviews and this comprehensive literature review.

3 Methodology

The methodology used to answer the research questions combines systematic literature review and bibliometric analysis. The review helped us understand the various research fields associated with Lightweight Consensus and the domains that require modified consensus.

3.1 Research Questions

The formulation of the following research questions paved the way for accomplishing the aforementioned contributions.

(1) **RQ1:** How does Lightweight Consensus contribute to the functionality of application-specific network domains?
(2) **RQ2:** What parameters define or classify the consensus as lightweight?
(3) **RQ3:** Which group of algorithms is modified to achieve characteristics of Lightweight Consensus?
(4) **RQ4:** Which types of blockchains benefit from Lightweight Consensus?

3.2 Data Collection

A search was done on Scopus, IEEE Xplore, and ARXIV database with the keywords "lightweight," "consensus," and "blockchain." Various databases were chosen to ensure that this review is a comprehensive one. The initial search was done without a year limit, only in English, and limited

[1]Internet of things, Peer-to-Peer, Lagre Scale Distribute Network, Partial Synchronous Network, Edge Computing, Mobile Network, Software Defined Networks.

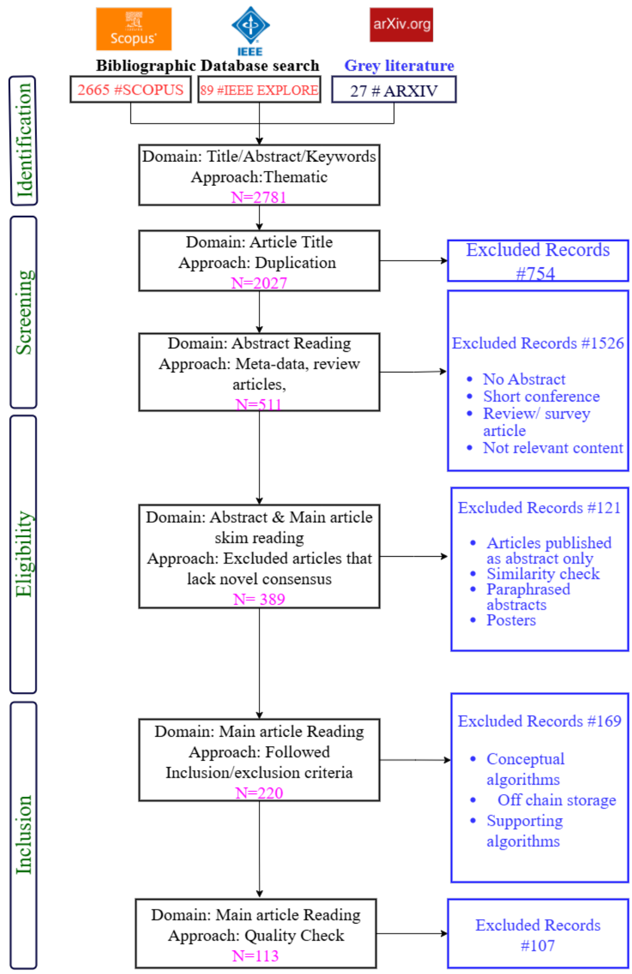


Fig. 1. Systematic review search strategy.

to journals, conference publications, and book chapters. The total number of results from the query was 2781. After the initial screening criteria, the total number of articles included was 511. The included papers were reduced to 389 based on the eligibility conditions. Finally, after the inclusion-exclusion criteria and quality check, the number included in the study was 113. Figure 1 details the eligibility criteria and the number of articles chosen at each stage.

3.3 Duplicate Elimination

This process eliminates duplicate records based on article titles in numerous databases. This is necessary to verify the validity and dependability of the review conclusions. Duplicate elimination was done by manually screening the entries. After eliminating duplicates, the remaining records were evaluated for eligibility and quality. This step reduced the total number of records to 511.

3.4 Abstract and Main Article Skim Reading

In this process, we read the abstract and eliminated short papers, review papers, and any articles that indicated no novel consensus algorithm.

Table 2. The Inclusion Criteria Used in this Study

	Inclusion criteria		Justification
IC1	All articles that mentioned improved blockchain architecture with consensus modification	1	Since the consensus is the crux of blockchain, any change to the architecture mostly involves modifying the consensus too.
IC2	All studies mentioned improved/lightweight/efficient consensus.	2	Since the focus of this study is Lightweight Consensus.
IC3	All studies which mention mining efficiency	3	Mining efficiency is improved by improving the consensus mechanism.
IC4	All studies which have analyzed the algorithm proposed	4	The analysis is important to understand what a Lightweight Consensus

Table 3. The Exclusion Criteria Used in this Study

	Exclusion Criteria		**Justification**
EC1	Articles which Improve storage by off-chain database	1	When the storage is off chain, scalability automatically increases and is not due to the consensus.
EC2	Articles which propose additional protocols to the existing consensus.	2	Additional protocols contribute to processing overhead.
EC3	Articles which propose consensus for any other network.	3	Our study's focus is on blockchain.
EC4	Conceptual proposal	4	Proposals that do not have any analysis to prove the effectiveness of the proposed consensus.

3.5 Inclusion and Exclusion Criteria

Inclusion and exclusion criteria are crucial components of a systematic review protocol because they help to define the study design characteristics of studies that will be included or excluded from the review. The inclusion and exclusion criteria define the scope of the systematic review. The inclusion criteria and the justification for choosing the specific criteria are outlined in Table 2.

The exclusion criteria help to exclude the articles not aligned with this study. Table 3 describes the exclusion criteria and the justification for choosing them.

3.6 Quality Assessment

The last step of the review process, and the most important one, is quality assessment. It entails assessing the methodological quality and relevance of the research included in the review to guarantee that the evidence-based conclusions are accurate and credible. We checked the assessment, and studies that reduced and proved the Lightweight Consensus were chosen.

3.7 Bibliometric Analysis

Bibliometric analysis is a quantitative analysis of academic literature and a method of science mapping. It is an application of bibliometrics, a quantitative analysis of written publications. It can show which researchers have published the most in a given field. As seen in Figure 2, most selected research was published as journal articles, followed by conference publications. There was only 1% published as book chapters. Figure 3 illustrates the increasing publications per year; these are only of the papers selected after the eligibility criteria check. Thus, indicating the growing interest and need for Lightweight Consensus in the blockchain. **Visualization of similarities** (**VOS**) viewer

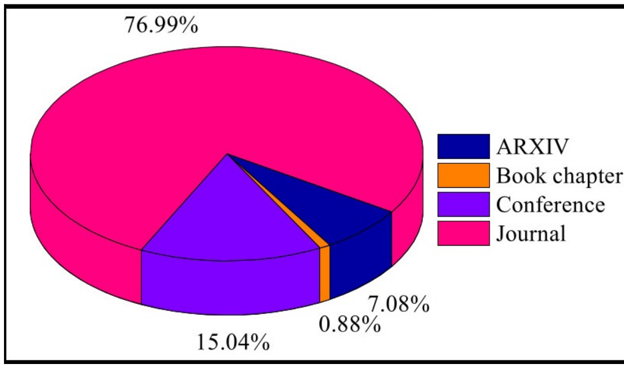


Fig. 2. Number of research publications.

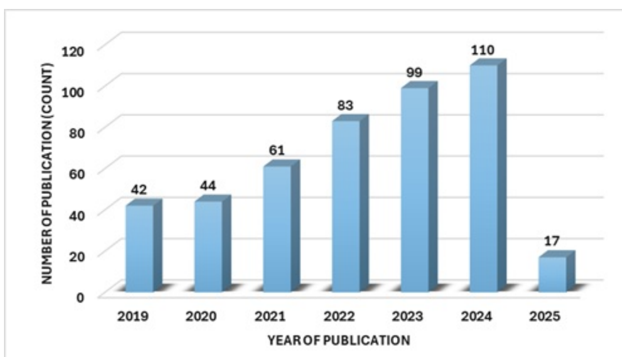


Fig. 3. Statistics of the year-wise publications of Lightweight Consensus.

[101] is used to analyze the articles. The primary objective of using a visualization tool is to equip readers with accessible information and better understand the trends in the Lightweight Consensus in blockchain. This method provides effective reasoning between the publication and associated research fields.

Keyword analysis. Figure 4, shows the network visualization of the various keywords associated with Lightweight Consensus algorithms. To obtain the keyword network, the co-occurrence analysis method was used. The threshold provided was full counting and a minimum of three occurrences of a keyword. Eight clusters were formed. Each keyword is associated with a circle; the large circles have more publications associated with the keywords. Two significant clusters, the consensus algorithm cluster and blockchain cluster, have links with all the other seven clusters. Both clusters have keywords associated with performance and application. An indication that researchers have focused on performance optimization and application development. The Distributed system cluster has a link to the **Practical Byzantine Fault Tolerance** (**PBFT**) cluster, as seen in Figure 5(a). In contrast, the edge computing cluster in Figure 5(b) has a link to the PoW cluster. Implying that the underlying network could play a role in the choice of consensus used.

3.8 Incorporating Recent Developments: Hand-Search Approach

The methodology of a systematic literature review encompasses all the steps represented in Figure 1. However, we have used a hand-search strategy as a supplementary approach to ensure completeness

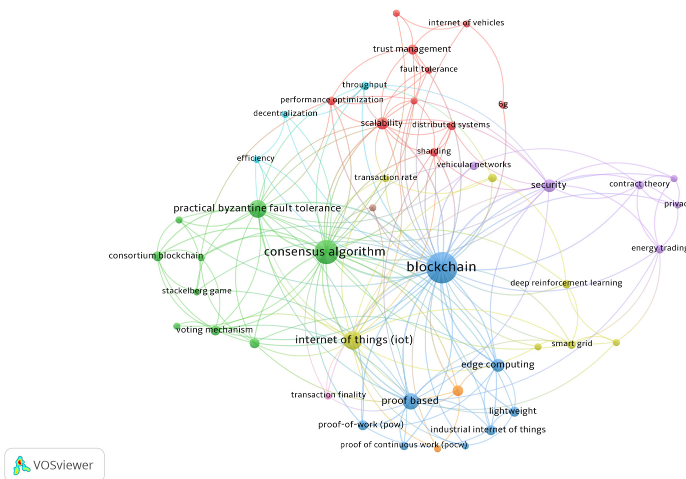
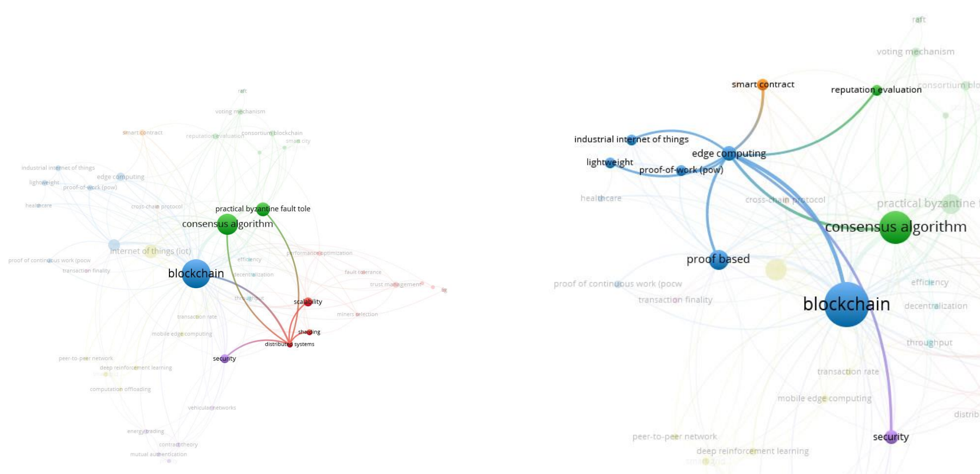


Fig. 4. Network visualization (VOSviewer) keyword analysis.



(a) Distributed system cluster (b) Edge computing cluster keywords.

Fig. 5. Clusters associated with networks.

and capture emerging trends. New research papers may not have been indexed at the time of the initial review or may have appeared in sources outside the originally selected databases. To bridge this gap, a targeted hand search was conducted in journals, databases, and archives related to lightweight blockchain. The selection criteria for these additional papers remain consistent with the original inclusion criteria in terms of scope and relevance. By incorporating these recent studies, we enhance the comprehensiveness of our review. With the hand search, an additional 14 papers were added. This is incorporated into this article.

4 Scope

This section defines the scope of this work. The evolution of consensus, with a timeline, is provided. Then the novel taxonomy is presented.

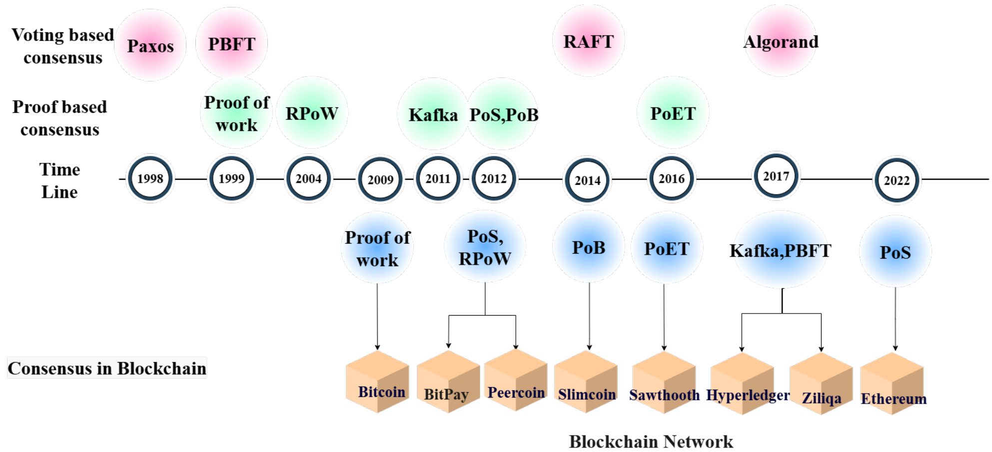


Fig. 6. Milestone of consensus algorithms.

4.1 Evolution of Consensus Algorithms

As blockchain evolved, so did the consensus algorithms. This section discusses the various classifications and evolution of consensus algorithms. Figure 6, depicts the milestones in the evolution of consensus and their application in various blockchain networks. The main line depicts the year in which the consensus was first proposed. They are grouped as voting-based and proof-based. Below, the timeline depicts the commercial usage of consensus in various blockchain networks.

Leslie Lamport proposed the PAXOS consensus method in 1998 to mitigate email spam [60]. PAXOS is a fault-tolerant consensus method that ensures that all nodes in a distributed system agree on a single value, even if some fail. Cynthia Dwork and Moni Naor [29] introduced PoW in 1993 to avoid denial-of-service attacks and email spam. Markus Jakobsson and Ari Juels [45] termed it "proof of work" in a study published in 1999. It was commercially used in Bitcoin in 2009. In the same year, the PBFT consensus algorithm was proposed by Miguel Castro and Barbara Liskov [15]. Hal Finney proposed the idea of **reusable PoW** (**RPoW**) in 2004. RPoW was an innovative way to obtain consensus in a distributed system without the use of a trusted central authority. Sunny King and Scott Nadal [55] developed Proof of Stake (PoS) in 2012 as an alternative to Bitcoin's energy-intensive PoW (PoW) mechanism. Iain Stewart presented Proof of Burn as an alternative to PoW and PoS in 2012. The Kafka consensus method is a distributed consensus algorithm that was initially presented in 2011 in a work titled Kafka: a Distributed Messaging System for Log Processing by Jay Kreps, Neha Narkhede, and Jun Rao [56]. Diego Ongaro and John Ousterhout [83] initially proposed the Raft consensus method in 2014. Raft is a consensus method based on leaders that is intended to be more intelligible than previous consensus algorithms, such as Paxos. Intel created the **Proof of Elapsed Time** (**PoET**) consensus mechanism in 2016. PoET is intended to be a less energy-intensive alternative to PoW. PoET is a cryptocurrency that is used in various commercial blockchain systems, including Hyperledger Sawtooth and Quorum. Silvio Micali et al. [19] invented Algorand, a consensus algorithm, in 2017. In 2022, Ethereum, one of the most prominent blockchain networks, transitioned from a PoW consensus process to PoS. This was a significant advancement for the network as it addressed some of the restrictions associated with PoW. The switch to PoS also allows Ethereum to deploy new scaling methods more effectively.

The use of consensus algorithms in commercial blockchains was much later than initially proposed. This undoubtedly reflects the need to properly investigate the need for blockchain and appropriately improve the consensus.

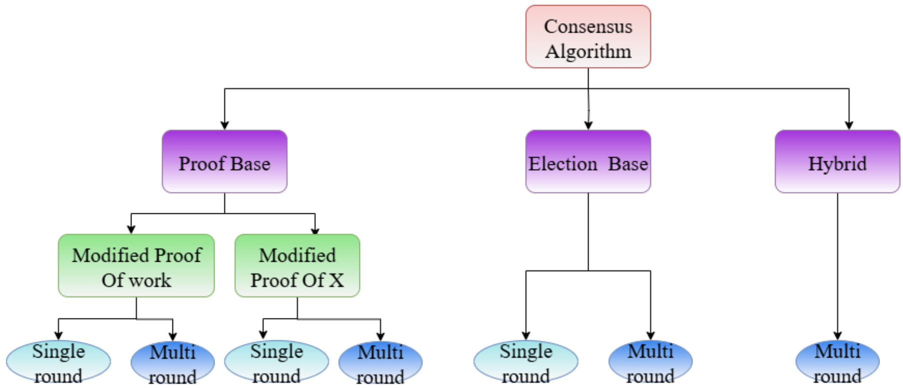


Fig. 7. Classification of lightweight algorithms.

4.2 Definition of a Lightweight Consensus

A Lightweight Consensus algorithm is a consensus mechanism designed to be efficient and resource-conserving, requiring minimal processing power, communication, and energy overhead. Lightweight Consensus algorithms differ in their configuration properties, performance properties, and implementation details. They are typically used in systems where resources are limited, allowing nodes to participate in the consensus process without demanding significant computational or energy resources. Section 6 discusses various parameters that help determine a Lightweight Consensus.

4.3 Classification of Lightweight Consensus Algorithm

This literature classifies the lightweight algorithms illustrated in Figure 7. The main purpose of improved consensus algorithms is to mitigate the problems of existing consensus and make it suitable for practical applications. The classification is as follows:

(1) **Proof-based:** The main problem with PoW is the high amount of energy it uses. The advantage is high scalability and security. There are two ways to improve the PoW: one is to redistribute the amount of work needed. The second is to change the work to some other quantifiable quality, like coin age. Hence, we further classify this category into modified PoW and modified proof of X, where X stands for any resource that must be spent to prove trustworthiness. From the selected literature, it is observed that to achieve better consensus performance, the authors have introduced rounds to verify and validate the blocks.
(2) **Voting based:** Voting-based consensus algorithms require all the participants to vote for the most trusted node, which will be responsible for confirming the block. Most voting-based algorithms are on a consortium blockchain and require a certain degree of trust before adding the nodes to the network.
(3) **Hybrid:** This type of consensus algorithm is a combination of proof-based and voting-based. These algorithms are usually multi-round consensus, with each round representing the respective type of consensus.

Table 4 compares the features of the different consensus categories. The various criteria chosen are

(1) Nodes join freely: This indicates that new nodes participate in the consensus hassle-free, with minimum delay.
(2) Decentralization: This describes whether the blockchain network is decentralized.
(3) Participating Nodes: Defines if all the nodes in the network participate in the consensus or only a group of selected nodes are permitted to partake.

Table 4. Comparison of Categories of Lightweight Consensus

Criterion	Proof-Based	Election-based	Hybrid
Node join freely	Yes	No	Nonminer nodes can join.
Decentralization	Yes	No	Miner election is centralised
Participating Nodes	Unlimited	Limited	Limited
Trust requirement	Low	High	High
Award	Huge	Minimal	Minimal

Table 5. Comparison of Single-Round and Multi-Round Mechanisms in Lightweight Consensus

Sr No	Properties	Single round	Multi-Round
1	Latency	Low Latency	Possible high
2	Easy of Implementation	Easy	Complex
3	Overhead	Reduced	Increased Overhead
4	Fault Tolerance	Low	High
5	Node Scalability	Low	High

(4) Trust requirement: The nodes need to provide a certain level of trust factors to participate in a consensus.
(5) Award: This is the reward for committing a block.

Furthermore, each category is sub-categorized as single-round or multi-round. A **round**, in a consensus algorithm, is a pre-defined time epoch, within which a set of transactions needs to be committed. At the end of a round, a new leader may be selected, or a new challenge may be initiated to select a new winner. In a single round, as the name suggests, in a given time epoch, the winner commits a set of transactions. In a multi-round system, in the first time epoch, a voting or some sort of selection algorithm is run to select a leader, and in the subsequent rounds, the leader commits the transaction. Following this, a new leader is selected at the end of a phase (time epoch of multiple rounds).

4.4 Examination of Lightweight Consensus Algorithm Classifications

Often, with limitations on resources like bandwidth and computing capacity, Lightweight Consensus methods are made to efficiently reach agreement across dispersed nodes. There exists a tradeoff in the conception and use of single- and multiround consensus procedures by analyzing their differences, in Table 5, on various performance parameters.

5 Domains Specific Challenges of Blockchain Consensus

This section discusses the various domains in which researchers felt the need for Lightweight Consensus algorithms. From the selected list of papers, we have narrowed it down to eight different application networks/domains that need an improved or Lightweight Consensus. The statistics are illustrated in Figure 8. As seen in the pie chart, the most Lightweight Consensus is published for applications that work on large-scale distributed networks, and the least work has addressed edge computing and mobile computing.

Under each domain, several specific applications are possible. Below is the list of the domains, their descriptions, and the sub-list of the respective applications.

5.1 Internet of Things (IoT)

An IoT architecture allows effortless incorporation of heterogeneous devices that perform diverse tasks ranging from sensing to **Machine-to-Machine communication** (**M2M**) [49]. Currently, IoT follows a centralized system approach, which exhibits flaws such as a common point of failure,

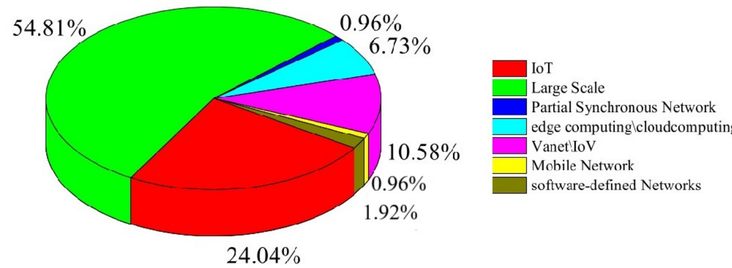


Fig. 8. Number of publications categorized into different application networks.

network bottlenecks, security, and scalability. A distributed system, such as blockchain, can solve the drawbacks of an IoT centralized network [105]. A Lightweight Consensus is required to improve blockchain performance in a resource-constrained IoT network, such as energy consumption, network communication, throughput, and storage. Table 6 lists all the studies that proposed a Lightweight Consensus for IoT as the underlay network. The goal of the consensus, the methodology used for verifying and validating the proposal, and possible research gaps are also presented in the same table. The goal of the Lightweight Consensus in IoT-based blockchains ranges from improving PBFT consensus to improving security and scalability. Zhendong Ai and Wei Cui [2] propose **Proof of transactions** (**PoT**), a hybrid multi-round consensus protocol that aims to achieve higher performance without losing security, fairness, and other features. In the first round, the node with the most valid transactions is considered the winner; in the subsequent round, all the peer nodes verify the block created by the winner. Jinwen Xi et al. [109] propose **Blockchain Consensus based on Global Reputation** (**BCGR**), which utilizes reputation information and timestamp from IoT devices to ensure the reliability of nodes while supporting node election and committee formation to enhance blockchain security. It is a single-round hybrid Voting-based consensus. Miao Du et al. in [27] highlight that the 3D-GHOST consensus mechanism improves security and network performance by ensuring that distributed peers can reach consensus on transaction history. This is achieved by using a **directed acyclic graph** (**DAG**) foundation composed of macroblocks, which are larger blocks that contain more transactions. The 3D-GHOST mechanism prioritizes macroblocks, which helps to maximize system throughput and network security.

5.2 Peer-to-Peer (P2P)

P2P networks are a type of decentralized network architecture in which peers, or individual devices, communicate without needing a centralized server or intermediary. P2P networks are becoming increasingly popular due to their ability to distribute large amounts of data quickly and efficiently without costly infrastructure. Some applications of blockchain technology use P2P networks to create a secure and transparent distributed ledger of transactions. There are several advantages to using P2P networks over traditional client-server architectures. They are less prone to failure because no single point of failure could bring the entire network down. They are also more scalable, as adding more nodes increases the network's capacity and speed. However, P2P networks have some disadvantages. One of the most difficult challenges is ensuring security and privacy in a decentralized network without a central authority to enforce rules or monitor activity. Overall, P2P networks have the potential to revolutionize the way we share and distribute information, but security and privacy must be carefully considered. A Lightweight Consensus attempts to solve some of the above-mentioned issues. Table 7 summarises the papers that have proposed a novel consensus in P2P networks. It also discusses the goal of the consensus, the methodology used to validate the consensus, the research gap or future works, if any, and the results obtained.

Table 6. Summary of Lightweight Consensus in IoT Networks

Sr. No	Consensus Name	Goal of Consensus	Methodology	Research Gap or Future work
1	Proof-of-Transactions (PoT) [2]	To handle huge transactions in the IoT network	Theoretical analysis and experiments to validate node election	Test in real-world scenarios
2	PBFT based on reputation value [39]	To increase the performance of BPFT	Nodes with higher reputation are prioritized	NA
3	multiple leaders PBFT [87]	To improve the performance of the PBFT	Ordering service using a discrete-time Markov chain, simulation and experimentation	Security analysis needs to be done
4	Byzantine Fault-Tolerant consensus algorithm based on Dynamic Permission Adjustment (DPA-PBFT) [69]	To ensure IoT data security through data consistency	Simulations based experimentations	Need to increase scalability
5	Hybrid Fuzzy Possibility C Means Clustering (HFPCM) [85]	To solve IoT issues with control authentication on trust	Simulation and experimentations, using c means	Needs to be enhanced by taking into account additional elements like interference analysis and packet reliability rate
6	Improved PBFT consensus mechanism based on reward and punishment strategy (SPBFT) [64]	To reduce the computational and communication overhead	Experiments and comparative analysis	Additional overhead is introduced for calculating behaviour score
7	Honesty-based Distributed Proof-of-Authority (HDPoA) [3]	To provide a secure and efficient consensus	PoW and PoA combine to provide security, accountability, and integrity	Energy efficiency is low
8	Distributed Time-based Consensus (DTC) [26]	To randomly select a block generator	Eliminate the need to solve a complex puzzle and clustering	Tested only on smart home senario
9	Improved consensus [66]	To improve the PBFT consensus	Uses identity-based secret sharing	Extension of the proposed consensus algorithm to support more operations and applications
10	Three-Dimensional Greedy Heaviest-Observed Sub-Tree (3D-GHOST) [27]	To improve security and network performance	Uses three-dimensional ledger architecture, unique data structure, and parallel workflows.	Open security issues need to be addressed
11	Proof-of-Trust (PoT) [123]	To build a scalable blockchain with high transaction-per-second (TPS)	Design and analysis on simulation.	Explore other applications of IoT
12	BCGR based on global reputation [109]	To improve trust	Experiments shows better in scalability and security.	Possible overhead due to time synchronizations.
13	LBlockchainE consensus [48]	To address security problems in the storage of sensor data	Docker simulations	Analysis on security and efficiency need to be done
14	Byzantine-based consensus mechanism (BCM) [38]	To improve scalability and throughput	Simulations	NA
15	Proof-of-computational closeness (PoCC) [89]	For miner selection and block creation	Uses resistance distance, to calculate computational closeness, this is evaluated by simulation analysis	Specifically for smart homes, can be expanded to other applications
16	Proof of Disease (PoD) [98]	To solve challenges faced in medical challenges	NA	It needs qualified medical miners and a cloud based server, which are costly resources
17	Proof of equivalent Work(PeW) [31]	To improve the reliability	All nodes need to perform equivalent work	Experiment on the different reward system in PeW
18	Credit Reinforce Byzantine Fault Tolerance (CRBFT) [20]	To improve PBFT	NA	Needs increase security
19	Proof-of-ENF [116]	To provide a secure and Lightweight Consensus system	Proof-of-concept prototype for leveraging Electrical Network Frequency (ENF) signals used as proof	validate on real-world scenario
20	Reputation-Driven Dynamic Node Consensus [47]	To improve the consensus and reliability of nodes	Evaluation and analysis of the technique, reputation score	use of machine learning techniques to predict node behavior and improve the accuracy of reputation calculation.
21	Lightweight-enhanced PoAh [51]	To reduce computational operations	Simulation analysis on Multi-zones-DAG blockchain	Ne ed a detailed explanation of the new consensus
22	CBCIoT [100]	To improve scalability for validation and verification IoT application	Simulation based experiment and analysis	Extended work on the security
23	Tree-Chain [25]	To increase speed and scalability in validator selection	Simulation based design and implementation	Study the impact of various attacks.
24	Blockchain Consensus Transform (BCT) [34]	To enhance efficiency and ensure fault tolerance	Experimentations and analysis of credit-based technique	Study the impact of network topologies, machine learning techniques, real-world IoT scenarios, in other applications.
25	Proof of Believability and Trust (PoBT) [10]	To provide a scalable secure and efficient Lightweight Consensus	Two step verification: trade verification and consensus formation	Future optimization

Table 7. Summary of Lightweight Consensus in P2P Networks

Sr. No	Consensus Name	Goal of Consensus	Research Gap or Future work	Results
1	Compute and Wait in PoW (CW-PoW) [53]	To reduce the energy consumption of PoW.	Specifically for cryptocurrency, delay in block finalization	Proposed an effective consensus for a generic public blockchain.
2	Delegated Proof of Stake with Downgrade (DDPoS) [120]	To improve the DPoS algorithm.	Security analysis for specific attacks.	An efficient, secure consensus that uses fewer resources
3	Proof-of-Lottery(PoL) [9]	To enhance the scalability of blockchain	No block rewards, No malicious nodes punishment.	A scalable consensus algorithm based on the lottery selection method.
4	URSA [88]	To improve storage and reduce forks	Requires real-world testing	Enhanced Nakamoto consensus.
5	Robust Round Robin (RRR) [1]	To optimize the performance of traditional consensus.	Designed for cryptocurrency applications, lacks an incentive mechanism.	An optimized algorithm that improves traditional consensus

The studies in this domain mostly focus on improving traditional consensus protocols. The application domain is specifically for cryptocurrency transactions. Mostefa Kara et al. [53] propose to **compute and wait in the PoW** (**CW-PoW**) consensus algorithm, which aims to optimize the energy consumption of POW. The ingenuity lies in a compute-and-wait mechanism. The nodes attempt to solve a cryptographic challenge and wait for the rest to catch up before going ahead to solve the next. The challenge complexity increases with each iteration and network power. Thus, ensuring constant block time results in lower energy consumption. Experimentations have shown an energy gain presentation of 16.63% when tested for 5 rounds. **Delegated Proof of Stake with Downgrade** (**DPoSD**), a consensus protocol proposed in [120], by authors Fan Yang et al., aims to prevent power concentration while ensuring security and efficiency. Nodes are designated as witness nodes and candidate nodes. The candidate is selected based on the highest votes and regularly rotated to avoid the accumulation of power. There is also a downgrading mechanism, which replaces malicious nodes. Sung-bin Lee et al. propose the **Proof-of-Lottery** (**PoL**) [9] consensus, which improves the scalability of blockchain networks based on **proof-of-stake** (**PoS**). The PoL algorithm employs a lottery-based mechanism to randomly select block producers from a pool of validators based on their stake, reducing the number of validators needed for block production and improving network scalability. The URSA [88] protocol aims to enhance the original Nakamoto consensus by a self-adaptive mechanism that adjusts the block size and difficulty level in response to network transaction volume changes. The article proposes a **Robust Round Robin** (**RRR**) protocol that addresses the shortcomings of **Proof-of-Work** (**PoW**) and PoS protocols, such as high energy consumption, centralization, and vulnerability to attacks. The RRR [1] protocol uses a deterministic leader selection mechanism based on a round-robin schedule, where each node in the network takes turns to be the leader and proposes a new block.

5.3 Large Scale Distributed Network (LSDN)

An LSDN is an architecture designed to support large-scale data processing and storage across multiple nodes or computers. An LSDN connects various computers to work as a single system, and each computer in the network is responsible for processing and storing a portion of the data. LSDNs are commonly used in cloud computing, data centers, and **content delivery networks** (**CDNs**). They can be designed to scale horizontally, meaning more nodes can be added to the network as needed to meet increased demand. LSDNs frequently use a central server or controller to manage the network for large-scale data processing and storage. Blockchains on LSDN suffer from various shortcomings [121]. The scalability to support large transactions is a huge issue. Verification and fairness of the consensus are also a concern for LSDN. The articles identified in this domain mainly focus on improving the consensus to overcome scalability and fault tolerance of the network while improving the consensus efficiency. Table 8, summarizes the various research done on the LSDN lightweight consensus.

Most of the papers selected for this study focus on LSDN, as these networks support a wide range of applications. Key use cases include supply chain management, smart grids, energy trading, crowdsensing applications, healthcare systems, and more. These applications rely heavily on the capacity of LSDN to handle large-scale data processing. However, when blockchain is introduced as a solution, traditional consensus mechanisms impose additional overhead in terms of energy consumption and resource usage. This highlights the need for Lightweight Consensus algorithms that can take advantage of LSDN and blockchain while minimizing energy and resource demands.

5.4 Partial Synchronous Network

In Partial Synchronous Networks, nodes are not fully synchronous. Yet, they have some level of timing coordination, allowing them to agree on a time frame to perform some operation or computation. A partially synchronous system is one where there is no global clock. In this type of network, nodes rely on message passing to coordinate their actions. There may be some uncertainty or variability in the communication and processing delays. One popular algorithm for coordinating actions in partially synchronous networks is the PAXOS algorithm, which is used for achieving consensus among a group of nodes. Leslie Lamport introduced the PAXOS algorithm in the study titled "The Part-Time Parliament" in 1998 [5]. Jan Hackfeld proposes a novel consensus Lisk-BFT algorithm [41]. It is a voting-based multi-round consensus explicitly designed for the Lisk ecosystem. The algorithm ensures that all validators agree on a single version of the blockchain, even in the presence of Byzantine faults. The blocks are proposed in a round robin method, in which each validator proposes one block at a time. When a validator offers a block, they provide two extra numbers in the block header: one for their vote and one for the overall number of votes from other validators. Following the proposal of a block, additional validators might vote on it by providing their vote integer in the following blocks. If more than two-thirds of the validators vote in favor of a specific block, that block is declared committed and is added to the blockchain. The Lisk-Bft method employs a tie-breaking technique to determine which block should be committed in case of a dispute. Testing the blockchain model of the network was done using the partially synchronous system model.

5.5 Edge Computing

Edge computing is a computing paradigm that involves computing and data storage closer to data sources, such as IoT devices or local edge servers. Edge computing can be utilized in **Vehicular Ad Hoc Networks** (**VANETs**) to provide real-time processing and analysis of network data. Edge computing, for example, may be used to analyze traffic. **Mobile edge computing** (**MEC**), or multi-access edge computing. The computational demands of using the PoW consensus algorithm in blockchain-enabled MEC can be a challenge for edge devices with limited computing resources. To address this, the article [76] proposes a novel consensus mechanism called **separate-proof-of-deep-learning** (**S-PoDL**) that uses a two-stage model to achieve computational tasks in a more energy-efficient manner. Simulations and analysis show that S-PoDL is more feasible and efficient for blockchain-enabled MEC applications while reducing the computing burden on edge devices. The number of malicious nodes in the blockchain network does not affect S-PoDL. Thus, it is more useful in real-world blockchain-enabled MEC networks with more dishonest nodes. In the

Table 8. Summary of Lightweight Consensus in LSDN Network

Sr No	Consensus	Goal of Consensus	Methodology	Research Gap and Future work
1	Green Proof-of-Work (Green-PoW) [62]	Reduce energy consumption of POW	Stochastic and qualitative analysis	No discussion of the runner-up nodes
2	Split Byzantine Fault Tolerance (SplitBFT) [79]	To improve trust of BFT	Experimental	Slight overhead added.
3	Trust-based PBFT (tPBFT) [99]	To improve the performance of alliance blockchains safely and efficiently	Theoretical analysis and experiments	More centralised
4	Feature grouping and Credit optimization Byzantine Fault Tolerance (FCBFT) [21]	To enhance PBFT	Experimental simulations	More centralized
5	Raft-based BPFT (R-PBFT) [63]	To improve network scalability	Experimental simulations	NA
6	New Byzantine Fault Tolerance (NBFT) [121]	High fault tolerance consensus	Experimental and analysis	Scalability needs to be improved.
7	IRD-IURM [129]	To minimize resource consumption and speed up block verification	Analysis and simulation	NA
8	Hierarchical PBFT [71]	To optimize the performance	The experimental results show shorten consensus total time.	Difficult to practically implement
9	Prosecutor [126]	To detect faulty behavior and ensure fairness in Byzantine servers	Penalization mechanism suppresses and marginalizes harmful assaults by Byzantine servers.	No analysis of the scalability
10	K-medoids cluster-based Proof of Action(KPoA) [102]	Improve the power efficiency and malicious node detection of PoA	Uses K-medoids clustering, and experimentations	Increase security
11	Scalable in-memory & shared-storage consensus [78]	Reduce resource consumption and make it suitable for HPC systems	Prototype implementation and evaluation	To include Parallel data integration
12	Fixed degree of decentralization Delegated Proof of Stake (FD-DPoS) [75]	To balance decentralization and efficiency in blockchain systems	Theoretical analysis and experimental simulation	Explore the use of machine learning techniques
13	Jointgraph [110]	To improve consensus time using DAG	Simulations	More power to supervisory nodes
14	SPBFT [130]	To reduce communication overhead.	Simulations	Lack of empirical validation
15	ipbft [65]	To achieve low latency	ipbft has shorter time and lower latency	NA
16	K-means Delegated Proof of Stake (K-DPoS) [73]	To reduce the degree power concentration in nodes	Simulations to test the use of K-means algorithm to select good nodes	Expand the application domain
17	Improved Raft [35]	To optimize the Raft consensus	Experimentations on Hyperledger	Analyse security and fault tolerance
18	Scalable Hierarchical Byzantine Fault Tolerance (SHBFT) [67]	To improve BPFT for consortium blockchain	Prototype implementation	NA
19	Proof of Descriptor [84]	To increase scalability by leaderless consensus	Lock-Free Transactional Transformation (LFTT)	NA
20	Proof of Value (POV) [24]	To provide equal awards	NA	Not clearly explained the PoV

(Continued)

Table 8. Continued

Sr No	Consensus	Goal of Consensus	Methodology	Research Gap and Future work
21	Credit-Delegated Byzantine Fault Tolerance (CDBFT) [106]	To improve PBFT	Simulations,using credit evaluation method	Lack of security analysis
22	Fast Byzantine Fault Tolerance (FastBFT) [72]	To improve PBFT, make it fast and scalable	Uses a novel message aggregation technique, which involves hardware with secret sharing	Hardware dependency
23	Proof-of-Contribution(PoC) [95]	To give users incentive to contribute to the consensus	Design and implementation in experimental setup	Explore other applications
24	Anonymous Proof of Location (A-PoL) [54]	To protect the identity of nodes, while using their location as proof	Game theory and optimization-based methods provide a stable solution for energy market	Improve scalability, integration of renewable energy sources, pricing mechanisms, and machine learning to optimize energy trading.
25	Voting-Based Decentralized Consensus (VDC) [96]	To achieve faster consensus process and better user fairness	Use verifiable random function (VRF)	NA
26	Proof-of-Behavior (PoB) [30]	To encourage honest behaviour and mitigate malicious nodes	Behavior-based incentive mechanism	NA
27	SPoCS [22]	To adapt the hybrid networks	Used Deep Reinforcement Learning (DRL) combined with Deep Q-Network (DDQN) optimization algorithm	NA
28	IPBFT [14]	TO enable PBFT to solve large throughput and achieve fast confirmation speed	NA	NA
29	Interference-based consensus [68]	To reduce system overhead and promote spectrum sharing	Transaction validation and interference-based consensus validate spectrum transactions to avoid interference.	Explore other applications
30	Improved distributed consensus [77]	To enhance the performance of consensus	NA	NA
31	Modified consensus [119]	To reduce malicious nodes	Experimentations	No proper comparison with existing consensus
32	[124] PoCW	To incentivize miners to store and provide access to data	Theoretical analysis and simulation results demonstrate that the proposed system has higher reliability than existing systems	Simulation or real-world testing missing
33	Hermes BFT [46]	To design a consensus to deal with slow primary node	Simulations and theoretical analysis	Security analysis not done
34	Delegated Proof of Reputation (DPoR) [44]	To ensure the security	A case study and numerical simulations	designed specifically for trading system
35	Modified Zyzzyva [94]	To improve scalability and fault tolerance	proposes the use of three consensus protocols	need of a trusted node
36	Delegated Proof of Reputation (DPoR) [4]	To reduce the computational expenditure of PoW	Simulations and theoretical analysis	Security analysis not done

(Continued)

Table 8. Continued

Sr No	Consensus	Goal of Consensus	Methodology	Research Gap and Future work
37	Dynamic reputation-based consensus mechanism, using DPBFT [13]	To improve PBFT for low-latency applications	experimental analysis, using reputation scaring	there is only one monitoring node.
38	Proof-of-Negotiation (PoN) [33]	To avoid miners being fixed.	trusted random selection algorithm and parallel multi-blocks creation	No security analysis
39	Difficulty-based Incentives for Problem Solving (DIPS) [86]	To Optimize consensus	Rewards blockchain miners for solving optimization problems	Test in other blockchain networks
40	Improved consensus [70]	To improve the security and efficiency	Mortgage model	Energy consumption not explored
41	Proof of Game (PoG) [58]	To provide a Lightweight Consensus mechanism for healthcare applications	Experimentations	Improve the efficiency and effectiveness
42	Improved DPoS [113]	To improve the efficiency and equity	Uses vague sets	additional complexity
43	Proof-of-Credit (PoC)-Voting-based Chain Finality (VCF) (PoC-VCF) [115]	To maintain the global blockchain state	Combination of theoretical analysis, design, and implementation	NA
44	Improved PBFT [125]	To improve scalability	Experimental analysis	No security analysis
45	ShardCons [57]	To improve scalability in public blockchains	Shards are determined using the K-means algorithm	Missing experimental analysis
46	Microchain [114]	To improve the reliability, security, and privacy of large-scale networks	Prototype Implementation	committee selection on large scale networks need to be practically verified, re-design of chain structures.

accounting queue technique, S-PoDL may generate several blocks in a single block creation cycle. Thus, reducing block time and improving the transaction rate. Also, the block validation criteria prevent energy theft and double-spending attacks.

In another work, [118] **Edge Computing Blockchain security Consensus Model** (**ECBCM**), a security consensus mechanism is proposed to solve edge computing-oriented blockchain security challenges. The use of a prestige-based approach guarantees fault tolerance. In case of an attack, a node replacement policy is in place. The **Credit score-based PBFT consensus** (**CSPBFT**) protocol is a consensus method proposed in Reference [103] to improve a blockchain system's long-term operating efficiency by simplifying the consistency protocol and successfully identifying and deleting rogue nodes. The node credit score method, which awards credit scores to nodes depending on their behavior, effectively identifies and excludes Byzantine nodes. Nodes with low credit ratings are deemed malevolent and are barred from participating in the consensus process. The credit scores and fraction of malicious nodes in CSPBFT decline continually while the system runs, successfully identifying and eradicating malicious nodes. [108] presents **Proof of Schedule** (**PoSch**), a new consensus technique that uses Stackelberg games to increase the acceptability of created schedules in cloud computing settings. Nevertheless, the authors admit that combining blockchain architecture and algorithms with the scheduling process to increase security is still a work in progress because the scheduling criterion is an open research subject. A summary of all the research work done in this domain is presented in Table S7 in the supplementary document.

Reference [81] proposes a new PoR Lightweight Consensus that addresses some of the drawbacks of current consensus methods like Delegated Proof of Stake. The novel PoR consensus mechanism aims to increase the security and effectiveness of the blockchain-based MEC system (DPoS). With enhancements in block verification and miner node selection, the PoR consensus process seeks to accomplish this goal. PoR chooses reliable miners for block generation and verification using a reputation score evaluation approach for miner node selection. Also uses block verification, PoR employs a lightweight technique that permits each miner to check with a different node just once throughout the consensus procedure. This saves network bandwidth and considerably lowers verification latency.

Reference [117] outlines a Lightweight Consensus mechanism to increase blockchain throughput and decrease the amount of transactions that need to be validated in new blocks. Each **cluster head** (**CH**) node in the blockchain must check the new block received from other nodes before connecting it to the chain as part of the consensus mechanism, which also entails confirming the signature of the block generator. The minimum time of the consensus cycle is almost twice as long as the longest end-to-end latency between nodes, and the default maximum for the consensus period is 5 minutes.

5.6 Mobile Network (MN)

MEC is a technology that enables computing resources and services to be hosted at the edge of the MN, closer to end-users. MEC can be used to offload computationally intensive tasks from mobile devices to nearby servers, reducing the energy consumption and processing power required by the device. In a blockchain-based MEC system, a single MEC server is located at a **base station** (**BS**) to handle computationally extensive data tasks offloaded from **end devices** (**EDs**). A blockchain network is deployed over the MEC system, where each ED acts as a blockchain miner. This allows for enhanced security and data integrity in the offloading process. Table 9 summarises the Lightweight Consensus in this network paradigm.

5.7 Software Defined Networks (SDN)

Because the control plane and data plane in conventional network topologies are closely connected, network hardware (such as routers and switches) is in charge of both forwarding data packets and deciding how to route them. SDN, in comparison, divides the data plane from the control plane. Network managers may operate the entire network from a single location since the control plane is centralized and programmable. This enables dynamic setup of network services as well as the more effective management of network resources. Generally, compared with conventional systems, SDN offers a more effective and adaptable method of managing networks.

Sujie Shao et al. [91]propose an optimized PBFT algorithm to improve the efficiency and security of the PBFT consensus. A trust evaluation method, a streamlined trust-based PBFT algorithm, and the use of node reputation as an election proof are among the proposed improvements. In order to reach an agreement on the system's status, nodes communicate with one another. The improved PBFT algorithm seeks to minimize any weaknesses brought on by malicious or malfunctioning nodes while offering quick and secure verification and confirmation of transactions inside a distributed system.

5.8 Unmanned Aerial Vehicle (UAV) Ad-Hoc Networks (UANET)

UAVs have gained popularity due to their speed, cost efficiency, and ease of use. However, they face limited computational efficiency and security challenges. Despite their benefits, UANETs face security concerns, including potential control signal spoofing, data integrity issues, and single points of failure due to reliance on centralized servers. A UANET enables multiple UAVs to collaborate and communicate with each other on a decentralized platform. UANETs face various issues, including limited scalability, low latency, data integrity risks, and various security issues. Blockchain technology offers a decentralized solution, providing security, data integrity, and privacy

Table 9. Summary of Lightweight Consensus in Mobile Computing

Sr. No	Consensus Name	Goal of Consensus	Methodology	Research Gap and Future work
1	Proof-of-Federated-Learning (PoFL) [6]	To improve efficiency and reliability for emergency message propagation	Theoretical and practical analysis of the use of federated learning and smart contract	can improve performance by including cross-layer information in the dataset
2	Score grouping BPFT (SG-PBFT) [112]	To improve efficiency and security of BPFT	Combination of theoretical and experimentation analyses of score grouping system	Optimize the node selection strategy
3	Efficient Delegated Proof-of-Stake (DPoS) [52]	To establish secure and traceable data sharing	Simulation base experiments of miner selection strategy	Improve scalability
4	PoR [18]	To improve computational efficiency and incentivize vehicle to participate	Simulation analysis of the reputation evaluation and block validation	Enhance security and scalability
5	Improved BPFT (PBFT) [40]	To impart secure message propagation	Simulation-based experiments to evaluate the proximity-based selection mechanism	Introduce credibility scoring module
6	Delegated proof-of-work (DPoW) [42]	To create a more robust network	Introduces delegate nodes to act on behalf of vehicles	Study the effect of various
7	Diffused BPFT (DPBFT) [16]	To reduce latency	theoretical analysis and comparison	explore the use of the consensus in other applications
8	Proof of Pseudonym [50]	To improve security and efficiency	Combination of experiments and analysis of the use of	extend the application of the
9	Consensus algorithm based on proof-of-work and PBFT [74]	To enhance reliability and efficiency	Simulation based consensus	Enhance the privacy protection level
10	Lightweight Consensus [128]	To enhance the consensus time	Combination fo various Byzantine technique	Enhance the accuracy with machine learning techniques
11	New consensus based on tip selection algorithm (TSA) [17]	To enhace efficiency	Simulations to analysis the efficiency of the algorithm	Extend beyond te proposed application
12	Byzantine Agreement (BA) algorithm for string consensus [90]	To enhance security	Simulations and analysis	Possible further optimization and improvement
13	Enhanced consensus algorithm based on PBFT [92]	To improve efficiency	Uses a dynamic reinforcement learning (DRL) approach to select consensus nodes.	Assessment in real-world scenarios

for UANETs. However, deploying traditional blockchain faces challenges like storage overhead, power consumption, and centralization risks [59].

A few researchers have proposed an improved Lightweight Consensus Specific for UANET. Zhoujie Wang et al. [107], have enhanced PBFT consensus by integrating a reputation-based evaluation system, selecting high-reputation UAVs as miners for improved efficiency, periodically updating the committee for scalability, and implementing weighted proposal voting to strengthen group decision-making in UANETs. Runqun Xiong et al. [111] propose a trust-based PBFT consensus mechanism for UANET blockchain, integrating UAV behavior-based trust evaluation to enhance reliability and efficiency. Trust scores, stored in a blockchain ledger, are influenced by positive behaviors and penalized for negative actions. UAVs monitor peers and can initiate accusation contracts for suspected misconduct, with trust values dynamically updated through consensus. **Lightweight and secure PBFT (LS-PBFT)** consensus proposed by Huihui Wang et al. [104] enhances PBFT by introducing pre-selection and group scoring mechanisms to improve consensus efficiency while reducing communication costs.

6 What Makes Consensus Lightweight?

There are various issues with the immediate adoption of state-of-the-art consensus (Refer: Section 1.1). We identified six key challenges that authors working on Lightweight Consensus try to overcome. The list of challenges and solutions proposed by the authors to address these issues is presented below.

6.1 Storage Scalability

One of the requirements of blockchain nodes is that all the participating nodes need to store the chain of blocks. Even though the transactions are hashed and stored, over time, in networks that have large transactions, storage does pose a challenge. The nodes participating in the consensus during the block confirmation need to have the chain of blocks to verify the parent hash before committing new blocks. Thus, participating nodes need to have greater storage capabilities. The solutions proposed by the authors are listed below:

- Edge node: Storing data in the edge nodes; these are nodes with higher resources, for example, servers. This is possible only in a private or consortium blockchain. There is an extra overhead for edge device authentication. The articles which address these are [97, 103].
- Sharding: This splits the chain into multiple sections, thus reducing the need to store lengthy chains. Transactions will be processed in parallel on each shard. Information about the shards needs to be broadcast and updated frequently to maintain security. However, sharding is difficult to implement in public blockchain networks. As it increases communication overhead and compromises security and privacy. The articles that use sharding are [30, 47, 57, 94, 94].
- Blockchain Compression: This is the use of various algorithms and data structures to reduce the data size after the chain has reached a specific length. [123] uses an authorized node to perform the compression to create a new genesis block; the node will extract the tree information from the last confirmed block and combine it with nonce, signature, and timestamp.
- Leaderless consensus: Leaderless consensus mechanisms like Proof of Descriptor [84] do not rely on a single validator to propose new blocks; instead, clients work together to complete transactions and assemble blocks. This approach allows parallel block building and publishing, improving performance, energy consumption, and being more resilient against attacks from malicious nodes.

6.2 Network Scalability

A primary concern with blockchain systems is network scalability, particularly as it pertains to consensus methods. The scalability challenges in a blockchain network become apparent as the number of users and transactions rises. With the decentralized nature of blockchain, where each node must process and validate each transaction, consensus-building might experience bottlenecks and delays. The consensus procedure's increased duration and computational cost constrain the network's performance. In order to increase the overall effectiveness and scalability of blockchain networks, efforts are being made to solve these difficulties using a variety of strategies, as listed below:

- Node grouping: The nodes are grouped based on location, a certain trust value, or hierarchy.
- Trust equity: The trust equity score is used to determine the trustworthiness of each node in the blockchain network. The trust equity score of each node in the network is calculated

by fusing and weighting the index data of high-quality and low-quality dimensions. This facilitates message communication between consensus nodes, enhances the Byzantine fault tolerance method, and supports more nodes [99, 103].
— Network fragmentation: To increase scalability and decrease latency, a big blockchain network may be fragmented into smaller sub-networks or pieces [63].
— Decentralization and democratization: The blockchain's decentralization and democratization principle ensures that decision-making authority is distributed among all participants, making the network more secure and transparent [121].

6.3 Node Trustworthiness

In network environments such as the IoT, where participants have the freedom to join and leave at will, ensuring security becomes a paramount concern. Blockchains deployed in such networks require an additional level of trustworthiness from the participating nodes. Presented below are several approaches that enable the maintenance of lightweight characteristics while attaining this desired level of node trustworthiness.

— Reputation: Reputation is the idea that a node is thought to be good or trustworthy based on its previous dealings. Throughout transactions, each entity's quality is assessed using a statistic called the transaction reputation value. It is described as an entity's normalized service quality as judged by its trading partner in a certain transaction [18, 28, 43, 47, 52, 81].
— **Trusted Execution Environment** (**TEEs**): TEEs offer an isolated and secure environment to run. Even from local attackers, TEEs safeguard the nodes' confidentiality and integrity [79].
— Trusted random selection and multi-block creation: The trusted random selection algorithm chooses trustworthy proposers and validators from a group of nodes in a blockchain network for a round of block generation. To increase the effectiveness of creating blocks in blockchain networks called parallel multi-block generation. Separating the mining team into groups can be implemented synchronously or asynchronously [33].
— Behavior score: The implementation of the reward and punishment strategy encourages nodes to act honorably and penalizes malevolent nodes [64].

6.4 Computational Intensity and Energy Inefficiency

Blockchain consensus protocols often require substantial computational resources, resulting in energy inefficiency and scalability challenges. To address these concerns, researchers have explored alternative mechanisms and metrics that aim at enhancing the efficiency and sustainability of blockchain systems. Below are key concepts and techniques related to these efforts:

— Several proof rounds: Multiple rounds are conducted to ensure trustworthiness.
— Selective mining: Using the energy used to mine one block to elect a few miners who will only work on that block in the future [62].
— **Verifiable Random Function** (**VRF**): A function that accepts an input and generates an output is known as a VRF. It is "verifiable" because there exist techniques that may be used to demonstrate that the output value is accurate. In other words, you can check the output's accuracy without being aware of the secret key used to create it. This is helpful when you want to make sure that a value was produced arbitrarily and without tampering, like in blockchain consensus systems. Setup, Prove and Verify make up the three key parts of the VRF [37].
— Utilization ratio: One important statistic for assessing the effectiveness of the Lightweight blockchain framework is the utilization ratio. A mathematical calculation that takes into account the number of nodes in the network, the consensus cycle, and the average speed of transactions generated by nodes per second can be used to determine the utilization ratio

of a blockchain network. You can change the consensus period or the number of cluster heads after clustering to alter the utilization ratio [117].

— Accounting queue: In this mechanism, N nodes whose model accuracy is the highest will be queued if N full nodes complete model validation. Thus, a maximum of N blocks can be generated in a block generation cycle.
— Block validation policy: Nodes are selected to mine and validate blocks based on their reputation score [3, 18].

6.5 Communication Overhead/Throughput

Communication overhead is incurred when blockchain is applied to dynamic networks. This, in turn, also affects the throughput and latency. Many of the solutions that address communication overhead also address throughput and latency [117]. **Diffused BPFT** (**DPBFT**) consensus [16] , reduces latency by performing the consensus process first in a small range around. It decomposes traditional PBFT into multiple sub-consensus areas, where participating nodes reach a primary consensus in a relatively small range and then are send to the entire network for further consensus. DPBFT also aims to reduce signaling costs and overall consensus by diffusing messages among nodes and introducing new steps such as Diffuse Preparation. Following are the ways to reduce communication overhead and increase throughput:

— Remove nonce: A nonce is to used to ensure randomness when hashing a block, but it is computationally taxing. Some studies do away with the nounce calculation [26].
— Improve block structure: This technique involves changing the block structure to accommodate for better security and throughput. [27] achieves better security by introducing macroblocks and microblocks, allowing for parallel processing eventually helps mitigating the risk of selfish mining. [23], which organises blocks into levels with a set width, implying that each round has a maximum number of new-generated blocks. Blocks in the current level point to blocks in the previous level, ensuring that the CoDAG grows into a well-connected channel. This topology enables high throughput between dispersed nodes.
— Alternative voting mechanism: Lightweight Consensus use alternate voting mechanism as a crucial part of the algorithm. It is used to address insufficient dynamic and low view switching. When a node quits or there is a Byzantine failure, the alternate node's message is broadcast instantly. This allows the alternate set to choose the next election action after replacing the elected trustworthy node with the consensus. Each node in the set that will take part in the election receives a message from the node that is taking part, letting them know they will be taking part [65].
— Stackelberg game: The Stackelberg game is a non-symmetric game where one player, called the leader, has a privileged position and makes decisions first, while the other players, followers, follow their actions. This game improves the approval of generated schedules in a cloud computing environment. The leader node generates a schedule and sends it to a follower node for verification and approval. The follower node can approve or reject the schedule based on its interests. This game allows for strategic decision-making by both parties involved in the transaction, ensuring that both parties act in their best interests while also considering how their actions will affect others. This model aims to improve security and efficiency in cloud scheduling [108].

6.6 Improve Performance of PBFT

PBFT is a widely used consensus mechanism for distributed systems, known for its fault-tolerance capabilities. However, as the scale and complexity of distributed networks grow, PBFT can face challenges such as communication overhead, inefficiency, and scalability limitations. To address these

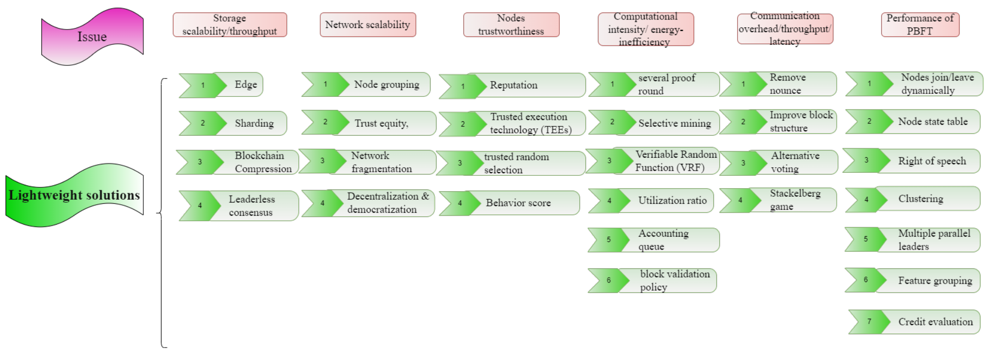


Fig. 9. Lightweight Consensus issues and solutions.

challenges, various techniques and optimizations have been proposed to enhance the performance, efficiency, and security of PBFT. Below is an exploration of these improvements.

— Nodes join/leave dynamically: The participating nodes can join or leave as they wish; this is done to ensure fault-tolerant behavior and scalability.
— Node state table: A state table is maintained for all the participating nodes.
— Right of speech/voting weights: Ensuring data consistency amongst non-peer nodes by dynamically altering their voting weights [69].
— Clustering is a technique used in **Wireless Sensor Networks** (**WSNs**) and IoT applications to group nodes into clusters based on proximity and energy levels to improve network efficiency and reduce energy consumption [85].
— Multiple leaders: For the PBFT ordering service, a number of concurrent leaders are used. Separate overlay networks, each with its own orderer and their interaction [87].
— Feature grouping/ score grouping: It entails categorizing nodes into groups based on their scores assigned during initialization. Higher-scoring nodes are assigned to the consensus nodes set and participate in the consensus process, whereas lower-scoring nodes are assigned to the candidate nodes set and only update their local states when a consensus is reached but do not participate in the consensus process temporarily. By reducing communication overhead and increasing throughput, this grouping mechanism helps to improve the efficiency of the consensus process [112].
— Credit evaluation/reputation value: Nodes with higher reputation are prioritized to reduce Byzantine nodes and improve consensus efficiency [20, 39].
— Identity-based secret sharing: This algorithm encrypts and distributes a secret key among multiple nodes in a distributed system using **identity-based encryption** (**IBE**). The key information is encapsulated as an authentication certificate to form a subkey, and the PBFT consensus algorithm uses the subkey as the voting basis for each node. The IBE public key encryption system uses a character string representing identity as the encryption public key, and the algorithm can be implemented using an elliptic curve and has adaptive selection ciphertext attack semantic security (IND-ID-CCA). This method enhances trust and increases security in distributed systems [66].

Figure 9 summarises the issues and the solutions offered by Lightweight Consensus for tackling the issues. There were a few drawback areas that were noticed during the study:

(1) For solutions involving node modifications, like node joining and leaving dynamically. This solution can be applicable to a small set of nodes, thus making scalability a challenge.
(2) For solutions that involved keeping track of nodes, like node state table or clustering, this reduced the decentralization property of blockchain.

(3) For solutions which offer a different voting mechanism, there is a possibility of concentration of power and super nodes getting voting rights for longer durations.

7 Properties of Lightweight Consensus

This section analyzes the work done and defines various parameters which help determine a Lightweight Consensus. From Section 4.3, it is evident that each class of algorithms has some unique features. When studying Lightweight Consensus, we found that the Various parameters can be divided according to the consensus properties as:

7.1 Configuration Properties

The parameters outlined below define the essential aspects of how nodes engage in the consensus process within a blockchain network. These settings encompass classification, agreement methods, network types, and node roles are described below:

(1) Classification: Based on the working of the consensus, they are classified according to the taxonomy provided in Section 4.3.
(2) Agreement: The method adopted by the consensus algorithm, which enables all the nodes to commit a set of transactions to the blockchain ledger.
(3) Blockchain network: This classification pertains to the type of blockchain network, which can be categorized as public, private, or consortium, each serving distinct purposes and governance structures.
(4) Node types: Each consensus algorithm defines different node roles, and this parameter indicates the distinct types of nodes involved in a consensus process, and their functions and responsibilities within the network.
(5) Block reward: For every successful block added to the chain, the winner receives incentives in various forms. This represents the incentive mechanism.
(6) Block time: Represents average block creation time.
(7) Fork avoiding mechanism: This is how the consensus algorithm avoids forks.

The details of the various configuration parameters of Lightweight Consensus in this study, have been outlined in Table S3 in the supplemental material.

7.2 Block Proposer Selection Properties

A crucial aspect of any consensus algorithm is establishing the criteria for block proposer selection, ensuring that all nodes have an equal opportunity to be chosen. This process must prioritize fairness, align costs with the value derived, and facilitate straightforward verification of the proposer's legitimacy. The various parameters associated with this property are listed as follows:

(1) Fairness: All nodes should have an equal chance of being selected as the block proposer.
(2) Investment: The cost of the block proposer selection process should be proportional to the value gained from it.
(3) Verification: It should be relatively simple to verify that the block proposer was legitimately selected.
(4) Honesty: Nodes participating in the consensus process should have a high probability of being honest.
(5) Termination: All honest nodes finally decide on a block. As this property affects the transaction commit time.
(6) Unforgeability: The block proposer selection process should be unforgeable, ensuring that no node can emulate fake attributes, and the block that is being agreed upon should be from a legitimate node

(7) Consensus Finality: A block once agreed upon and appended to the digital ledger, is not removed any time later.
(8) Security: The system should be resilient against common attacks on reputation systems and should not subvert the fundamental security guarantees of the blockchain.
(9) Decentralization: The consensus protocol should not be quasi-centralized by abandoning the decentralization property of the blockchain.

The various block selector parameters of Lightweight Consensus are detailed in Table S6 in the supplemental material.

7.3 Performance Properties

The following points outline the key performance parameters of a Lightweight Consensus algorithm, which are critical for assessing its efficiency and effectiveness in blockchain systems. These parameters include communication complexity, scalability, propagation of rounds, and transaction throughput, each playing a significant role in optimizing the consensus process

(1) Communication complexity: The time for a block to be committed. Reducing communication complexity is crucial for improving blockchain-based systems' performance and scalability.
(2) Scalability: This refers to the ability of the algorithm to process multiple transactions. High scalability indicates the ability to process multiple transactions.
(3) Propagation of rounds: The work done in one round is carried forward, giving an advantage to the nodes that will participat in the next round.
(4) Transaction throughput: This is the ratio of the total number of transactions and the time takes to confirm a block.
(5) Grouping: This property defines, wheter the nodes are grouped into specific types of nodes with special functions, like, cluster heads, or validators and edge nodes.

The details of the various performance parameters of Lightweight Consensus in this study have been outlined in Table S4 in the supplemental material.

8 Variations of Lightweight Consensus Across Different Domains

The use of blockchain in a variety of application domains, including MNs, **software-defined networks (SDNs)**, edge computing networks, VANETs, and IoT, calls for an enhanced consensus algorithm that is adapted to the particulars of each environment. Each domain has its specific drawbacks when traditional blockchain consensus mechanisms are implemented. To overcome these limitations, Lightweight Consensus algorithms have been proposed. As LSDN networks process vast amount of data, the modified consensus, which enhanced latency and scalability, will be most apt for this kind of network. A proof-based consensus with only a single round of leader selection would be highly beneficial in these applications.

For IoT, the consensus algorithm needs to be lightweight and energy-efficient, as IoT devices are resource-constrained in terms of power and computational capacity. The type of Lightweight Consensus most suited of IoT applications are, the PBFT based consenus algorithms, which fall into the election based classification of Lightweight Consensus.

VANETs and MNs are highly decentralised and the network topology keeps on changing. Various nodes leave and join the network. The applications which work on these networks, need high security with high throughput and low latency. The best suited Lightweight Consensus algorithms would be hybrid algorithms, which process transactions with low latency and high throughput. P2P networks needs increased security and scalability. Multiround proof based consenus algorithm, suits the needs of a P2P network applications. In edge computing, the consensus must address scalability issues, enabling rapid transaction validation across distributed edge nodes with minimal

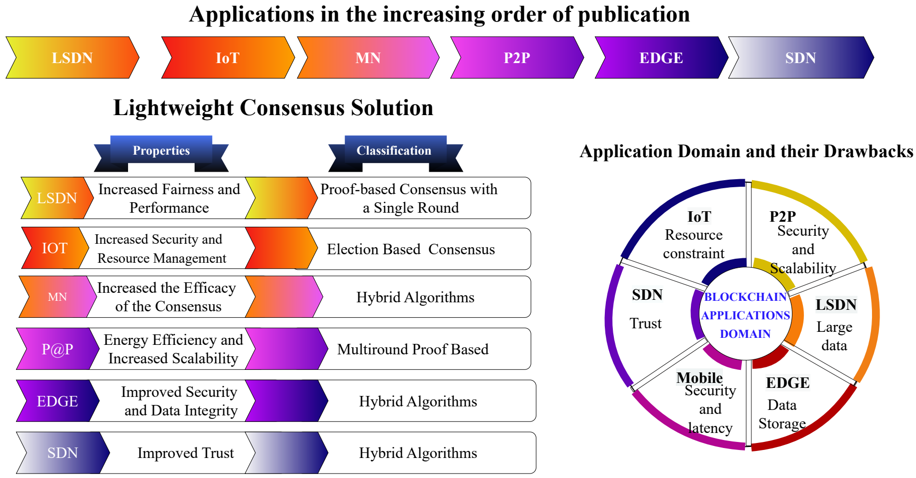


Fig. 10. Difference in Lightweight Consensus in different domains.

central control. SDNs, which rely on the centralization of network control, require a consensus mechanism that balances decentralization and centralized oversight, optimizing both security and performance. Any type of consensus mechanism with alternative storage would fit well for SDN applications.

Figure 10 summarizes the application domains, their associated drawbacks, the need for modified algorithms that can offer tailored solutions, and the best suited algorithm according to the classification.

9 Disscussion

This section attempts to answer the research questions posed in Section 3.1. This section discusses the difference between a Lightweight Consensus and a traditional consensus.

9.1 Conjecture to Research Question 1

Modern-day networks process huge amounts of data in a very short span of time. Distributed networks face an additional challenge of storage. Many studies propose off-chain storage as a solution. However, this compromises the decentralized feature of blockchain.

It is conclusive from Table 7 that in P2P networks 60% of the studies focus on improving the traditional consensus algorithm. Other parameters that are attempted to be improved are (i) Energy Efficiency, (ii) Scalability, and (iii) Storage In partial synchronous networks, the focus was on ensuring consensus in the presence of malicious nodes (Refer: Section 5.4).

In edge computing networks 76% of the papers focused on enhancing integrity and security (Refer: Table S7). The integrity of the miners and transaction validity was enhanced. Few studies also explored the integrity of the consensus algorithm in the presence of malicious Byzantine Nodes. Other studies enhance mining efficiency and resource allocation.

The studies in IoT networks proposed various Lightweight Consensus protocols that deal with (i) Scalability, (ii) Node Trustworthiness, (iii) Improving PBFT, (iv) Security, and (v) Reduce Resource Consumption.

Around 25% of the papers focused on improving scalability. The rest were about equally addressed. In large-scale networks, around 21% of the studies focused on improving fault tolerance and enhancing the PBFT algorithm. The other improved areas were scalability and security (Refer:

Table S7). The focus on improvement of efficiency in MN blockchain consensus was around 46% of the papers. The second area of improvement was security (Refer: Table 9).

From the above discussion, we conjecture that the underlying network to which the blockchain application is meant to cater has to be considered while choosing a consensus. With a focus on security, scalability, resource optimization, and fault tolerance.

9.2 Conjecture to Research Question 2

A lightweight consensus mechanism is optimized for efficiency, scalability, and security while minimizing computational complexity, communication overhead, and energy consumption. These properties can be grouped into three broad categories: Configuration Properties, Block Proposer Selection, and Performance Properties, as outlined in Section 7. In this subsection, we analyze the existing research and examine how various authors have incorporated a combination of these properties in their proposed lightweight consensus mechanisms. Tables S3, S4 and S6 highlight that a lightweight consensus desirably has a set of properties discussed as follows.

(1) Lightweight Consensus mechanisms often employ election-based or hybrid approaches rather than proof-based systems, enabling rapid block finalization with fewer propagation rounds, refer to Table S3 (from supplementary document) Configuration properties of Consensus.
(2) Block proposer selection mechanisms in Lightweight Consensus prioritize fairness, requiring low investment and enabling quick verification of legitimacy while ensuring security through reputation or stake-based models.
(3) Unlike PoW-based approaches, lightweight consensus avoids high energy consumption and computational costs by leveraging trust scores, voting mechanisms, or minimal proof requirements. These mechanisms typically operate in consortium or private networks, where predefined node roles (e.g., primary-replica, consensus nodes, validators) allow efficient governance. Additionally, they integrate fork-avoidance techniques (e.g., longest ledger, reputation-based finality) to ensure stability and reduce unnecessary network overhead.
(4) Security remains a priority, with resistance to attacks such as Sybil and double-spending, making lightweight consensus mechanisms an ideal choice for scalable and resource-efficient blockchain applications.

However, there is no single proposed solution that achieves all the desirable properties. This can be observed from our findings that there is a tradeoff between some of the properties, as Table S3, helps identify that hybrid protocols have comparatively higher block time. This indicates that multiple rounds of consensus lead to a certain amount of delay in committing blocks. Table S4 shows the multi-round consensus protocols have higher scalability and throughput. However, the drawback is that they have higher overhead. Those consensuses that target reducing communication complexity incur low scalability. Depending on the underlying network and the application, one must choose the best-suited consensus.

9.3 Conjecture to Research Question 3

The consensus agreement method can be classified into vote-based or proof-based. From Table S3, 90% of the studies have proposed a voting-based agreement method. Traditionally, byzantine fault-tolerant consensus follows a voting-based agreement. Also from Section 6.6, it is evident that a substantial amount of research is towards improving fault-tolerant-based consensus. We concur that many studies are motivated to improve fault-tolerant consensus due to its inherent feature of less energy consumption. However, BFT protocols face challenges related to scalability, communication overhead, and time sensitivity, which can make them vulnerable to **denial-of-service** (**DoS**) attacks.

To address these limitations, modified BFT-based consensus mechanisms introduce optimizations such as reducing the number of communication rounds, implementing leader election strategies to minimize redundant message exchanges, and adopting threshold cryptographic techniques to improve security without excessive computational costs. Additionally, some approaches use committee-based selection to limit the number of participating nodes in the consensus process, reducing overall latency and overhead.

Table S4 indicates that 80% of modified consensus have achieved considerable scalability while maintaining a moderate throughput. In particular, multi-round consensus modifications, despite achieving high scalability and throughput, incurred additional overhead due to increased message complexity. However, these tradeoffs highlight the ongoing research focus on optimizing communication efficiency while preserving the security and resilience of BFT-based consensus mechanisms.

9.4 Conjecture to Research Question 4

There are several sorts of blockchains that serve distinct objectives in the domain of application-based blockchains. Consortium blockchains, public blockchains and private are prevalent variants with unique properties.

From Table S3, most studies focus on consortium and public blockchains. A consortium blockchain is a network controlled by a set of organizations or nodes rather than being available to the public, as is a public blockchain. Only trustworthy nodes, often pre-selected organizations or nodes, can participate in the mining process and maintain the consensus mechanism in a consortium blockchain. The capacity of consortium blockchains to provide alternative functionality to various nodes is one of their primary features. Unlike public blockchains, which require all nodes to engage in the same set of operations, consortium blockchains allow various nodes to have varied roles and responsibilities. This flexibility allows nodes that are not participating in the mining process to join or leave the network at any time without interfering with the blockchain's general functionality. This feature improves the blockchain's scalability by allowing for dynamic participation and more effective resource allocation.

Furthermore, consensus time is reduced because consortium blockchains only contain trustworthy nodes in the mining process. The time it takes for the network to agree on the legitimacy of a transaction and add it to the blockchain is referred to as consensus latency. Consortium blockchains can achieve speedier consensus by limiting participation to trustworthy nodes because the nodes are already recognized and trusted by the consortium members. This expedited consensus procedure reduces latency and enhances the blockchain network's overall performance.

9.5 How does Lightweight Consensus Differ from Traditional Consensus

Compared with conventional or traditional consensus methods, Lightweight Consensus algorithms are typically more effective and require fewer computational resources. Nevertheless, there is not a standard criterion yet to distinguish between systems that are lightweight and those that are not.

10 Future Scope

While this study has tried to include most of the parameters and properties of Lightweight Consensus, there are still areas to be researched. Future research focusing on these aspects could provide valuable insights and practical advancements for both scholars and practitioners.

(1) Security: The security of consensus is a vast and multifaceted topic, and the inclusion of this topic in this study would stem from the research questions attempting to be answered. Future research could delve deeper into how lightweight consensus protocols can be fortified against a range of security threats. This would lead to the development of more resilient distributed systems, especially in IoT networks.

(2) Authentication for consensus: Authentication is verifying a node's identity. Some consensus needs nodes to be authenticated before participating in the consensus. Future studies could explore how lightweight authentication mechanisms can be effectively integrated without compromising efficiency.
(3) Non-repudiation: Non-repudiation is the ability to prove that a transaction took place. Investigating how lightweight consensus algorithms can incorporate robust non-repudiation mechanisms would be valuable for applications requiring strong audit trails, such as financial transactions and legal agreements.
(4) Attack: Future research could focus on the resilience of lightweight consensus mechanisms against various attacks, including Sybil attacks, **Denial of Service** (**DoS**), block withholding attacks, and pool hopping attacks. A particularly promising area of study is network partitioning attacks [32], which can disrupt communication between nodes and undermine voting-based consensus protocols. Exploring defense mechanisms and detection strategies could help readers understand how to build more fault-tolerant distributed systems
(5) Privacy: privacy is an important aspect to the blockchain participant. Data privacy must also be ensured after the block has been committed. Future studies could investigate advanced cryptographic techniques and privacy-preserving mechanisms within Lightweight Consensus frameworks.
(6) Real-time deployment: Most of the Lightweight Consensus proposals are conceptual or simulator-based; there is a need for real-time deployment. Future research should focus on real-world deployment and testing under practical conditions. This would provide practitioners with valuable insights into the feasibility, scalability, and operational challenges of deploying lightweight consensus mechanisms in live networks.

10.1 How will Consensus Algorithms be in the Future?

After an in-depth analysis of the reviewed research papers, the current trend highlights efforts to address specific shortcomings within each application domain. Looking ahead, key advancements are expected to focus on enhancing lightweight consensus mechanisms and integrating artificial intelligence (AI) to analyze the behavior of participating nodes and intelligently select a leader. This leader selection process would be based on several critical factors:

(1) Increasing Use of Trust and Reputation-Based Mechanisms: The studies reviewed indicate a growing emphasis on incorporating trust scores and reputation chains to enhance fairness and validation. This shift could lead to more adaptable consensus algorithms, particularly suitable for IoT environments where resource constraints are significant.
(2) Hybrid Systems for Improved Adaptability: Future consensus mechanisms are likely to leverage hybrid models that combine the strengths of multiple algorithms. This approach would promote fairness, consensus finality, and adaptability across diverse applications.
(3) Randomization for Fairness and Security: The use of random number generation and random hash values will be increasingly important for ensuring fairness and enhancing security, making consensus outcomes less predictable and more resistant to manipulation.

The future of consensus mechanisms will likely prioritize energy efficiency and resource awareness, with factors such as battery levels influencing participation in the consensus process. Additionally, there will be a heightened focus on advanced fault tolerance mechanisms to strengthen security without compromising operational efficiency. A significant development on the horizon is the adoption of hardware-enhanced security, relying on secure hardware components to facilitate consensus. The use of TEE and secure enclaves is expected to become more widespread, bolstering system integrity.

Another promising enhancement involves node clustering, where each cluster designates a leader responsible for committing transactions to a sidechain. This sidechain would periodically synchronize with the main blockchain, enhancing scalability and processing efficiency. Moreover, incorporating frequent authentication algorithms would help verify participating nodes, ensuring robust security and reliability.

To address the growing challenge of managing large volumes of data, future consensus algorithms could implement transaction filtering and distribute data across multiple storage locations. Coupled with advanced cryptographic techniques, this approach would enhance both data security and storage efficiency. Additionally, the introduction of multi-layered consensus mechanisms presents a promising avenue, allowing consensus processes to be optimized at various layers depending on the nature and sensitivity of the data.

11 Conclusion

The major goal of this study is to look at Lightweight Consensus in the context of blockchain. Compared with standard blockchain systems, the term "Lightweight Consensus" refers to creating consensus methods or protocols that are more efficient and need fewer computing resources. These lightweight alternatives attempt to overcome traditional blockchain systems' constraints and issues, such as scalability, energy consumption, and transaction throughput.

This study reviews 127 published publications to offer an overview of the status of research in the topic. The investigation found that a bulk of proposed solutions were intended to solve the difficulties that traditional blockchain systems face.

The study cites proof-based techniques and Byzantine fault tolerance as the most common ways for obtaining Lightweight Consensus. PoS or proof-of-authority consensus systems require participants to provide evidence of their stake or authority in order to validate transactions and protect the blockchain. Byzantine fault tolerance, on the other hand, is concerned with reaching consensus in the presence of malicious or malfunctioning nodes in the network. Furthermore, the research highlighted many barriers to Lightweight Consensus adoption, including the danger of centralization, difficulties in creating a fair distribution of rewards, and vulnerability to assaults.

The key outcomes of this work are:

(1) A novel taxonomy of Lightweight Consensus, based on the agreement method and number of rounds.
(2) A timeline of the evolution of consensus and its application in blockchain.
(3) A parametric analysis of the characteristics of Lightweight Consensus.
(4) An understanding of the current issues of consensus and the various solutions to them.

This study provides a concise documentation of consensus algorithms in the blockchain. With an emphasis on the various improvements to the existing consensus to be applied for interested researchers.

Acknowledgments

We acknowledge the support of the School of Computer and Engineering, Manipal Institute of Technology (MIT), Manipal Academy of Higher Education, Manipal, for providing an enabling research environment throughout this work. We also thank Vidya Kamath, whose earlier systematic literature review (SLR) provided the initial direction and impetus for undertaking this study.

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Received 30 May 2023; revised 22 February 2025; accepted 5 September 2025